

# Monetary Policy and Aggregate Demand

Macroeconomics B  
Lecture 4 (Chapter 17)

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## Where We Are

In the previous lectures you built the two main components of private demand:

- ▶ **Investment:** firms invest when the shadow value of capital is high relative to installation cost, and housing investment rises when Tobin's  $q$  is above 1.
- ▶ **Consumption:** households smooth consumption over time, but current income, wealth, interest rates, and borrowing constraints all matter.
- ▶ **Common lesson:** both  $C$  and  $I$  react to the **real interest rate**.

Today you combine these building blocks into a theory of **aggregate demand** and link it to **monetary policy**.

**Question:** how does a central bank move output by moving interest rates?

# Roadmap

1. **Why demand matters in the short run**
2. **Goods market equilibrium and the IS curve**
3. **Money market and the IS-LM benchmark**
4. **Monetary policy and the Taylor rule**
5. **From IS-LM to the AD curve**
6. **Unconventional monetary policy**

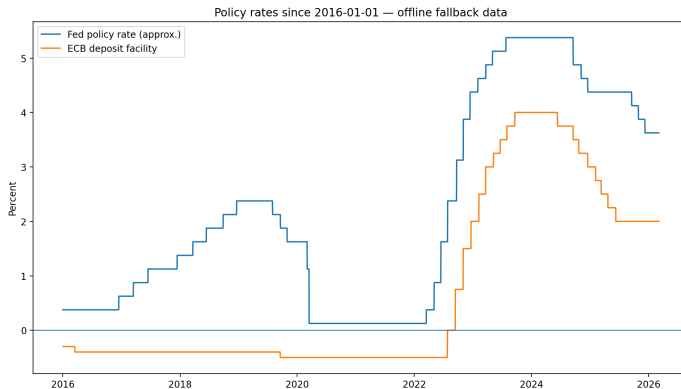
# What You Will Be Able to Do After This Lecture

After this lecture, **you** will be able to:

- ▶ derive a benchmark **IS curve** from an exact linear goods-market model and interpret the multiplier,
- ▶ use the **IS-LM diagram** to predict the effects of monetary, fiscal, and demand shocks on  $Y$  and  $r$ ,
- ▶ explain why modern central banks target an **interest rate** rather than a money stock,
- ▶ apply the **Taylor rule** and explain the economic roles of the parameters  $h$  and  $\beta$ ,
- ▶ derive and interpret the **AD curve**,
- ▶ distinguish between movements *along* the AD curve and **shifts** of the AD curve.

# Why Does This Matter? From Very Low Rates to Rapid Tightening

The ECB and the Fed kept policy rates very low for years, then tightened sharply once inflation surged.



Source: Based on FRED series EFFR and ECBDFR.

Policy rates moved from the lower bound to a sharp tightening cycle; the chapter asks how these moves pass through to the borrowing costs that drive consumption and investment.

# Motivation: Three Links We Need to Understand

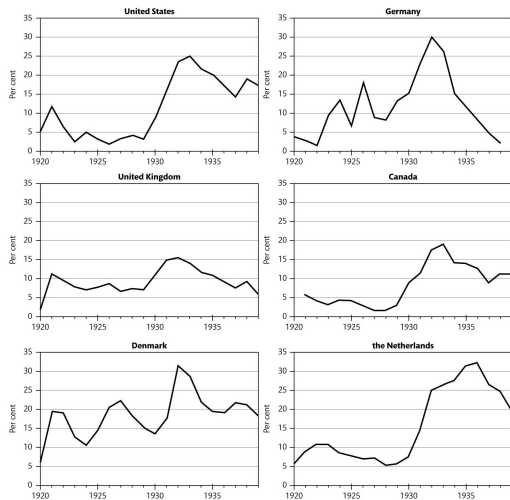
To explain short-run aggregate demand, you need three links:

1. **Goods market block:** how  $C + I + G$  determines output in the short run.
2. **Policy block:** how the central bank sets the short-term nominal interest rate.
3. **Transmission block:** how the policy rate affects the real rate relevant for households and firms.

Once these pieces are combined, you get the chapter's central result:

higher inflation  $\Rightarrow$  tighter policy  $\Rightarrow$  higher  $r \Rightarrow$  lower demand

# Keynes, the Classics, and the Great Depression



Source: Sørensen & Whitta-Jacobsen, Figure 17.1. Unemployment rates in six countries during the Great Depression. The persistence of high unemployment is the historical puzzle a demand-side model tries to explain.

## Classical view:

- ▶ output is mainly pinned down by supply,
- ▶ prices and wages adjust,
- ▶ long slumps should be self-correcting.

## Keynesian challenge:

- ▶ unemployment stayed high for years,
- ▶ output did not quickly return to normal,
- ▶ weak demand can keep the economy below  $\bar{Y}$ .

# From the Historical Puzzle to the Goods Market

The historical puzzle showed us that output can fall sharply even when an economy's productive capacity has not disappeared.

The Keynesian starting point is the closed-economy goods market:

$$Y = C + I + G$$

- ▶  $Y$ : output and income,
- ▶  $C$ : private consumption,
- ▶  $I$ : private investment,
- ▶  $G$ : government demand for goods and services.

The short-run perspective is:

**firms adjust output to match planned demand.**

This chapter asks:

1. What determines aggregate demand,  $C + I + G$ ?
2. How does monetary policy move that demand?



# Derivation of the IS Curve

**Goal:** turn the goods-market condition

$$Y = C + I + G$$

into a relation between output  $Y$  and the real interest rate  $r$ .

- ▶ Start from simple behavioral equations for consumption, taxes, and investment.
- ▶ Insert them into goods-market equilibrium.
- ▶ Collect the endogenous income feedback.
- ▶ Read the resulting IS curve as the goods-market equilibrium schedule.

# Private Demand: Consumption and Investment

We describe private demand with a simple linear setup:

$$C = C_0 + cY^d, \quad Y^d = Y - T, \quad T = T_0 + tY, \quad I = I_0 - br.$$

- ▶  $C_0$ : autonomous consumption, including confidence and wealth effects,
- ▶  $c$ : marginal propensity to consume out of disposable income,
- ▶  $T_0$ : autonomous taxes,
- ▶  $t$ : marginal tax rate,
- ▶  $I_0$ : autonomous investment,
- ▶  $b$ : sensitivity of investment to the real interest rate.

The richer demand channels from Chapters 15 and 16 enter through the same objects: autonomous private demand, taxes, and the real interest rate.

## Step 1: Insert Private Demand into the Goods Market

Start from goods-market equilibrium:

$$Y = C + I + G.$$

Insert consumption and investment:

$$Y = [C_0 + c(Y - T)] + [I_0 - br] + G.$$

Now insert the tax rule:

$$T = T_0 + tY.$$

This gives:

$$Y = C_0 + c(Y - T_0 - tY) + I_0 - br + G.$$

The term  $c(Y - T_0 - tY)$  captures the feedback from income to consumption: higher output raises income, higher income raises disposable income, and this raises consumption.

Derivation

## Step 2: Collect the Terms in Output

Rewrite the previous equation:

$$Y = C_0 - cT_0 + c(1 - t)Y + I_0 - br + G.$$

The term  $c(1 - t)Y$  is endogenous demand created by income itself.

Move this term to the left-hand side:

$$Y - c(1 - t)Y = C_0 - cT_0 + I_0 - br + G.$$

Factor out  $Y$ :

$$[1 - c(1 - t)] Y = C_0 - cT_0 + I_0 - br + G.$$

## Step 3: Solve for Equilibrium Output

Divide by  $1 - c(1 - t)$ :

$$Y = \frac{1}{1 - c(1 - t)} (C_0 - cT_0 + I_0 - br + G)$$

Define the multiplier:

$$m \equiv \frac{1}{1 - c(1 - t)} > 1.$$

Then:

$$Y = m(C_0 - cT_0 + I_0 - br + G).$$

The multiplier is larger when households spend a larger share of extra disposable income. It is smaller when the tax rate is higher, because taxes dampen the income-consumption feedback.

## Step 4: Write the IS Curve

Group autonomous private demand as:

$$A \equiv C_0 + I_0.$$

Then equilibrium output is:

$$Y = mA + mG - mcT_0 - mbr.$$

Define:

$$\alpha_1 \equiv m, \quad \alpha_2 \equiv mb, \quad \alpha_3 \equiv mc.$$

So:

$$Y = \alpha_1 G - \alpha_2 r - \alpha_3 T_0 + mA.$$

This is the IS curve: all combinations of  $Y$  and  $r$  that clear the goods market.

Full Derivation

## Step 5: Move from Levels to Gaps

Start from the level-form IS curve:

$$Y = \alpha_1 G - \alpha_2 r - \alpha_3 T_0 + mA.$$

Define the reference point using the same equation:

$$\bar{Y} = \alpha_1 \bar{G} - \alpha_2 \bar{r} - \alpha_3 \bar{T}_0 + m\bar{A}.$$

Subtract the reference equation from the actual equation:

$$Y - \bar{Y} = \alpha_1 (G - \bar{G}) - \alpha_2 (r - \bar{r}) - \alpha_3 (T_0 - \bar{T}_0) + m(A - \bar{A}).$$

Define the autonomous demand shock:

$$v \equiv m(A - \bar{A}).$$

## Reading the IS Curve

Write the IS curve in deviations from a reference point:

$$Y - \bar{Y} = \alpha_1(G - \bar{G}) - \alpha_2(r - \bar{r}) - \alpha_3(T_0 - \bar{T}_0) + v, \quad v \equiv m(A - \bar{A}).$$

The economic interpretation is direct:

- ▶ Higher government spending raises demand:

$$G \uparrow \Rightarrow Y \uparrow.$$

- ▶ Higher autonomous taxes reduce disposable income and consumption:

$$T_0 \uparrow \Rightarrow C \downarrow \Rightarrow Y \downarrow.$$

- ▶ A higher real interest rate reduces investment:

$$r \uparrow \Rightarrow I \downarrow \Rightarrow Y \downarrow.$$



# Why the Real Interest Rate Matters for Demand

In the simple linear setup, the real interest rate lowers demand through investment:

$$I = I_0 - br.$$

In richer demand models, the negative effect of  $r$  also comes from the earlier chapters:

## Consumption

- ▶ intertemporal substitution,
- ▶ lower wealth from asset-price declines,
- ▶ tighter cash flow for borrowers.

## Investment

- ▶ higher discount rate lowers  $q$ ,
- ▶ lower asset prices reduce firm value,
- ▶ higher user cost reduces housing demand.

Hence:

$$r \uparrow \Rightarrow C \downarrow, I \downarrow \Rightarrow Y \downarrow$$

# The LM Curve

**Goal:** add a second equilibrium condition so the model determines both output and the interest rate.

- ▶ The IS curve alone tells us demand for each possible real interest rate.
- ▶ The traditional LM curve adds money-market equilibrium.
- ▶ The intersection gives the short-run equilibrium pair  $(Y, r)$  or  $(Y, i)$ .

This benchmark is useful before we replace the money-market block with an interest-rate policy rule.

# Why the IS Curve Is Not Yet a Complete Model

The IS curve tells us how output depends on the real interest rate:

$$Y - \bar{Y} = \alpha_1(G - \bar{G}) - \alpha_2(r - \bar{r}) - \alpha_3(T_0 - \bar{T}_0) + v.$$

But it does not tell us where the interest rate comes from.

To determine output, we need a second relation:

- ▶ In the traditional IS-LM model, the interest rate comes from money-market equilibrium.
- ▶ In the modern New Keynesian model, the interest rate comes from the central bank's policy rule.

So the next step is to explain the traditional LM curve, and then why modern models usually replace it with an interest-rate rule.

# The LM Curve: Money-Market Equilibrium

LM stands for **liquidity preference** and **money supply**.

Money demand is:

$$\frac{M^d}{P} = L(Y, i), \quad L_Y > 0, \quad L_i < 0.$$

Money-market equilibrium requires:

$$\frac{M}{P} = L(Y, i).$$

- ▶ Higher income raises transaction demand for money.
- ▶ A higher nominal interest rate makes holding money less attractive.
- ▶ If real money supply,  $M/P$ , is fixed, then higher  $Y$  requires higher  $i$ .

Therefore the LM curve slopes upward in  $(Y, i)$  space:

$Y \uparrow \Rightarrow$  money demand rises  $\Rightarrow i \uparrow \Rightarrow$  money demand falls back to supply.

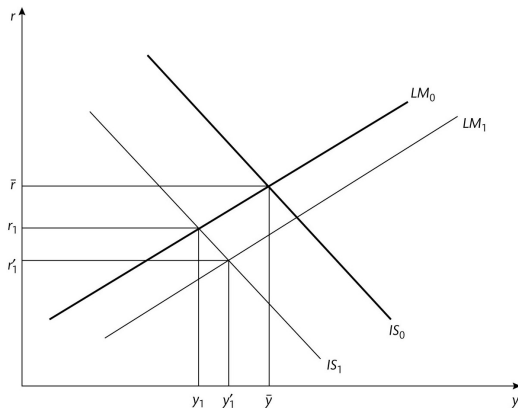
# The Full IS-LM Model

**Goal:** combine goods-market equilibrium and money-market equilibrium in one diagram.

IS: goods market clears      and      LM: money market clears.

- ▶ Shocks to spending move the IS curve.
- ▶ Shocks to money supply, money demand, or policy implementation move the LM curve.
- ▶ The intersection gives the model's prediction for output and the interest rate.

# Putting IS and LM Together



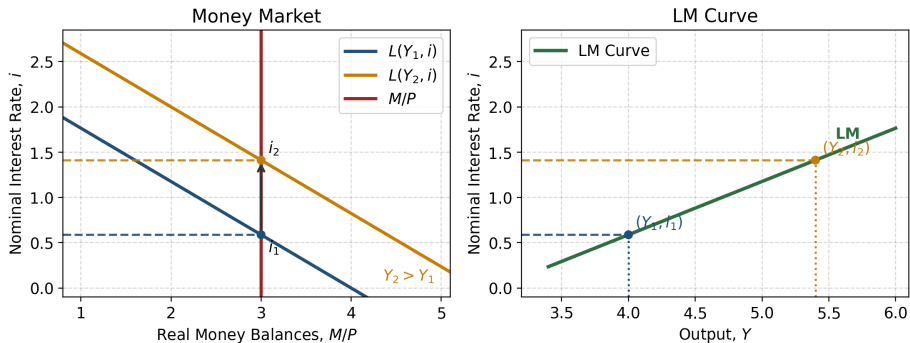
Source: Sørensen & Whitta-Jacobsen, Figure 17.2. The IS-LM model with shifts in IS and LM.

- ▶ The IS curve describes goods-market equilibrium.
- ▶ The LM curve describes money-market equilibrium.
- ▶ Their intersection determines output and the interest rate.

This picture is useful for intuition, but modern central banks rarely target the money stock directly.

# Graphical Derivation of the LM Curve

Graphical Derivation of the LM Curve



**Mechanism:** higher output raises money demand. With real money supply fixed, money-market equilibrium requires a higher interest rate. Connecting the implied equilibrium points gives the upward-sloping LM curve.

# The Taylor Rule

**Goal:** replace the traditional LM curve with a rule for how the central bank sets the policy rate.

- ▶ Modern central banks usually choose an interest rate, not a money stock.
- ▶ The policy rate anchors a horizontal policy line in an IS-policy diagram.
- ▶ The Taylor rule explains how this rate responds to inflation and the output gap.



# How Monetary Policy Works in Practice

A modern central bank does not move the money stock directly. It steers a single **policy rate**, typically the rate it pays on reserves or charges on lending to banks.

## Mechanics:

- ▶ The central bank announces a target for the short-term policy rate.
- ▶ It supplies or absorbs the reserves that banks need so the market rate stays close to the target.
- ▶ Other rates, on bank loans, mortgages, and bonds, then adjust through arbitrage.

## Three transmission channels to spending:

- ▶ **Borrowing-cost channel:** a higher policy rate raises the real cost of credit, so  $C$  and  $I$  fall.
- ▶ **Asset-price and wealth channel:** higher rates lower bond, equity, and house prices, reducing household wealth.
- ▶ **Expectations channel:** the path of expected future rates moves longer-term yields and shapes the borrowing rates that actually matter.

## Why We Replace LM with a Policy Rule

Today the central bank usually targets a short-term policy interest rate, not a money stock.

- ▶ Money demand is unstable, so fixing  $M$  does not deliver a stable interest rate.
- ▶ In practice, the central bank picks  $i$  and lets reserves adjust.
- ▶ Modern macro models therefore replace the upward-sloping LM curve with a **horizontal line at the chosen policy rate**.

The modern question is therefore:

How does the central bank choose  $i$ , and how does  $i$  affect the real rate  $r$ ?

This motivates replacing the LM curve with a monetary policy rule, for example a Taylor rule.

# The Taylor Rule: A Policy-Rate Equation

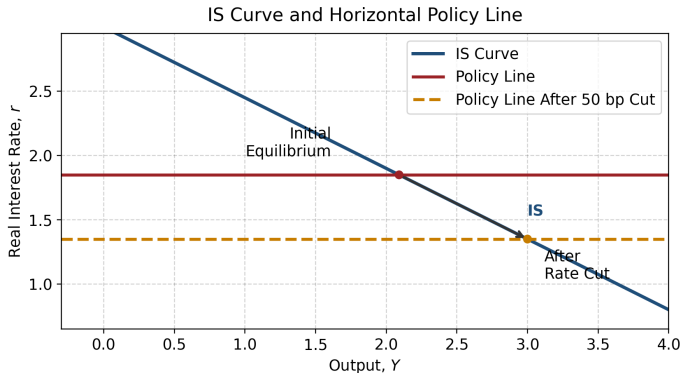
A simple interest-rate rule is:

$$i_t^p = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + \beta(Y_t - \bar{Y}) \quad h, \beta > 0$$

- ▶  $\bar{r} + \pi_{t+1}^e$ : neutral nominal policy rate,
- ▶  $h(\pi_t - \pi^*)$ : stronger policy response when inflation is above target,
- ▶  $\beta(Y_t - \bar{Y})$ : tighter policy when output is above its structural level.

This is a **rule-based** description of how monetary policy reacts to the state of the economy.

# Replacing LM with a Policy Line



- ▶ With an interest-rate target, the central bank supplies reserves to keep the policy rate at its chosen level.
- ▶ In the IS diagram this appears as a **horizontal policy line**.
- ▶ A rate cut shifts the policy line down; the economy moves along the IS curve to higher output.

# From the Policy Rate to the Relevant Real Rate

The policy rate is not always the rate relevant for private spending. Let

$$i_t = i_t^p + \hat{\rho}_t$$

denote the market rate that matters for households and firms, where  $\hat{\rho}_t$  is a credit spread or wedge. The exact **ex ante** real rate is

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e}$$

For the later AD algebra, the textbook uses the close approximation

$$r_t \approx \bar{r} + h(\pi_t - \pi^*) + \beta(Y_t - \bar{Y}) + \hat{\rho}_t$$

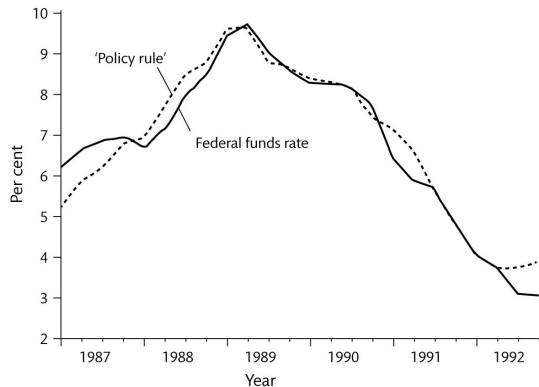
- ▶  $\hat{\rho}_t > 0$  means private borrowing costs are above the policy rate.
- ▶ This wedge will matter a lot in crisis periods and for unconventional policy.

## What Do the Taylor Parameters $h$ and $\beta$ Do?

| Parameter | Direct effect                             | Implication for the AD block                                              |
|-----------|-------------------------------------------|---------------------------------------------------------------------------|
| $h$       | stronger response to inflation deviations | makes monetary policy more inflation-focused and, later, makes AD flatter |
| $\beta$   | stronger response to the output gap       | dampens demand shocks on impact, but also makes the AD curve steeper      |

**Mechanism:**  $h$  changes how aggressively inflation moves the real rate, while  $\beta$  changes how strongly the central bank leans against demand itself.

# Taylor-Type Policy in the Data



Source: Sørensen & Whitta-Jacobsen, Figure 17.4, based on Taylor (1993).

A Taylor-type rule captures the medium-run logic of policy, even though actual decisions can deviate in exceptional periods.

- ▶ Taylor's original evidence showed that the US policy rate moved broadly with a simple rule.
- ▶ The fit is not exact, and it should not be exact, because central banks face shocks, uncertainty, and financial-stability concerns.
- ▶ A rule is therefore best seen as a **benchmark**, not a mechanical formula.

# Policy Lab: Predict the Effects

Before we move from the policy-line diagram to the AD curve, three short questions check the mechanics.

## Setup:

- ▶ Closed economy, diagram in  $(Y, r)$  space.
- ▶ The IS curve slopes down.
- ▶ Monetary policy is a horizontal policy line at the chosen real interest rate.
- ▶ Fiscal and investment shocks shift IS; policy-rate changes shift the horizontal line.

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## Policy Lab Q1: Government Spending

**Question:** The government increases  $G$ . The central bank keeps the policy rate unchanged.

In the IS-policy diagram, what happens?

- (A) The IS curve shifts right;  $Y$  rises and  $r$  is unchanged.
- (B) The IS curve shifts right;  $Y$  rises and  $r$  rises.
- (C) The policy line shifts down;  $Y$  rises and  $r$  falls.
- (D) Both curves shift left;  $Y$  falls.

## Policy Lab Q2: Pessimistic Firms

**Question:** Firms become pessimistic about future profitability and reduce autonomous investment  $I_0$ . The central bank keeps the policy rate unchanged.

In the IS-policy diagram, what happens?

- (A) The IS curve shifts left;  $Y$  falls and  $r$  is unchanged.
- (B) The IS curve shifts left;  $Y$  falls and  $r$  rises.
- (C) The policy line shifts up;  $Y$  falls and  $r$  rises.
- (D) Nothing changes, because investment will recover automatically.

## Policy Lab Q3: ECB Rate Cut

**Question:** The ECB cuts its policy rate by 50 basis points. Fiscal policy and autonomous demand are unchanged.

In the IS-policy diagram, what happens?

- (A) The policy line shifts down; the economy moves along IS, so  $Y$  rises and  $r$  falls.
- (B) The IS curve shifts right;  $Y$  rises and  $r$  rises.
- (C) The policy line shifts up;  $Y$  falls and  $r$  rises.
- (D) Only  $r$  falls;  $Y$  stays at  $\bar{Y}$  because output is supply-determined.

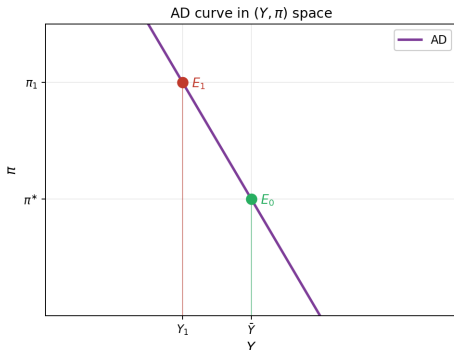
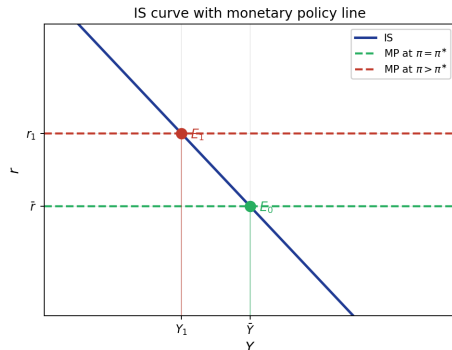
# From IS-LM to the AD Curve

**Goal:** translate the interest-rate mechanism into a demand curve in  $(Y, \pi)$  space.

- ▶ The IS curve maps the real interest rate into output.
- ▶ The Taylor rule maps inflation into the real interest rate.
- ▶ Combining both maps inflation into aggregate demand.

That combined relation is the AD curve.

# From IS-LM to AD: the Bridge



For each  $\pi$  the central bank sets  $r$ , and the IS curve gives  $Y$ . The mapping  $(Y, \pi)$  is the AD curve.

**Left:** for each inflation rate, the Taylor rule sets a real rate and a horizontal policy line.

Calibration:  $\alpha_2 = 1.5$ ,  $h = 0.5$ ,  $\beta = 0$ . Each policy line in the left panel corresponds to one point in the right panel.

**Right:** the resulting pairs  $(Y, \pi)$  form the downward-sloping AD curve.

## Combining IS with the Policy Rule

Take the IS curve and the real-rate rule:

$$\begin{aligned}Y - \bar{Y} &= \alpha_1(G - \bar{G}) - \alpha_2(r - \bar{r}) - \alpha_3(T_0 - \bar{T}_0) + v, \\r - \bar{r} &= h(\pi - \pi^*) + \beta(Y - \bar{Y}) + \hat{\rho}.\end{aligned}$$

Substitute the second into the first and collect  $Y$  terms:

$$(1 + \alpha_2\beta)(Y - \bar{Y}) = \alpha_1(G - \bar{G}) - \alpha_3(T_0 - \bar{T}_0) + v - \alpha_2h(\pi - \pi^*) - \alpha_2\hat{\rho}.$$

Define

$$a \equiv \frac{\alpha_2 h}{1 + \alpha_2 \beta} > 0, \quad Z \equiv \frac{\alpha_1(G - \bar{G}) - \alpha_3(T_0 - \bar{T}_0) + v - \alpha_2\hat{\rho}}{1 + \alpha_2\beta}.$$

The demand side becomes:

$$Y - \bar{Y} = Z - a(\pi - \pi^*)$$

Full derivation

# What Shifts the AD Curve, and What Does Not

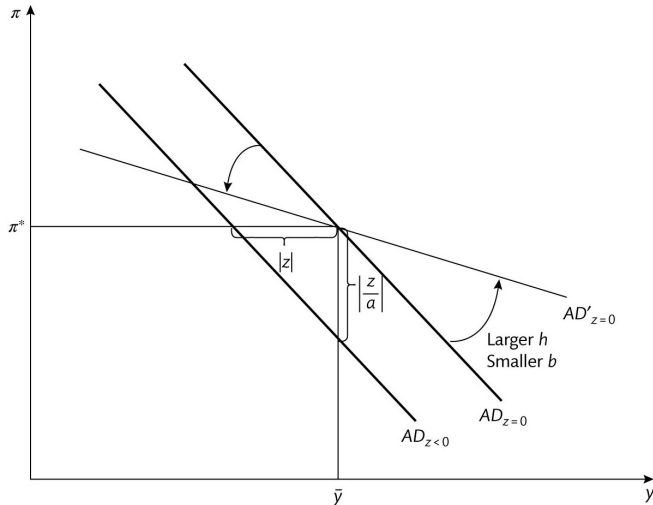
Recall:

$$Y - \bar{Y} = Z - a(\pi - \pi^*), \quad Z = \frac{\alpha_1(G - \bar{G}) - \alpha_3(T_0 - \bar{T}_0) + v - \alpha_2\hat{p}}{1 + \alpha_2\beta}.$$

| Shift AD right ( $Z \uparrow$ )      | Shift AD left ( $Z \downarrow$ )       | Movement along AD                              |
|--------------------------------------|----------------------------------------|------------------------------------------------|
| $G \uparrow$ (fiscal expansion)      | $G \downarrow$ (austerity)             | change in $\pi$<br>(CB reacts via Taylor rule) |
| $T_0 \downarrow$ (tax cut)           | $T_0 \uparrow$ (tax hike)              |                                                |
| $v \uparrow$ (autonomous demand)     | $v \downarrow$ (negative demand shock) |                                                |
| $\hat{p} \downarrow$ (lower spreads) | $\hat{p} \uparrow$ (financial stress)  |                                                |

**Key distinction:** a change in inflation *moves* the economy along a given AD curve. A change in  $G$ ,  $T_0$ ,  $v$ , or  $\hat{p}$  *shifts* the entire AD curve.

# The AD Curve in a $(Y, \pi)$ Diagram



Source: Sørensen & Whitta-Jacobsen, Figure 17.10. The AD curve for alternative values of  $Z$ ,  $h$ , and  $\beta$ . A positive demand shock shifts AD to the right, while a more inflation-responsive central bank makes AD flatter.



# Unconventional Monetary Policy: The Toolkit

The standard channel is the policy rate. When that channel weakens, for example because the lower bound binds or because credit spreads widen ( $\hat{p}_t$  rises), central banks reach for additional tools:

- ▶ **Forward guidance:** communicate the future path of the policy rate to anchor longer-term yields.
- ▶ **Liquidity operations:** provide cheap funding to banks (e.g., the ECB's TLTROs) to compress the wedge  $\hat{p}_t$ .
- ▶ **Asset purchases (QE):** buy government and corporate bonds to lower long-term yields and ease financial conditions.

In the language of our model, unconventional policy aims to move the spread  $\hat{p}_t$  and to reinforce the link between the policy rate and the real cost of borrowing for households and firms.

# Back to the Opening Hook: 2022–25 Through the Model

The chapter's mechanism reads:

$$\pi \uparrow \Rightarrow i^P \uparrow \Rightarrow r \uparrow \Rightarrow C, I \downarrow \Rightarrow Y \downarrow.$$

Interpreting the recent ECB cycle through the AD framework:

- ▶ Higher inflation pushed the economy along the AD curve, because the Taylor rule raised  $r$ .
- ▶ Where credit conditions also tightened,  $\hat{\rho} > 0$  added a leftward shift of AD.
- ▶ As inflation eased and the ECB started cutting from June 2024, the economy began moving back down along AD.

**Class question:** Was the 2022–23 slowdown in the euro area mostly a movement *along* AD, a leftward *shift* of AD, or both? What evidence would you point to?

## Notebook: `L4_monetary_policy_aggregate_demand.ipynb`

A companion notebook accompanies this lecture.

It lets you reproduce the IS-LM diagram, the policy rule, and the AD curve numerically, and explore how policy and demand shocks move them.

Sections are marked as **core** or **optional**. Use it after class to rehearse the diagrams and shock experiments.

# Takeaways

- ▶ In the short run, **aggregate demand** can determine output, which is why Keynes focused on  $Y = C + I + G$ .
- ▶ The **IS curve** summarizes goods market equilibrium:  
$$Y - \bar{Y} = \alpha_1(G - \bar{G}) - \alpha_2(r - \bar{r}) - \alpha_3(T_0 - \bar{T}_0) + v.$$
- ▶ The **IS-LM diagram** is useful for analyzing fiscal, monetary, and demand shocks.
- ▶ Modern central banks target an **interest rate**, not a money stock, and the **Taylor rule** is the standard benchmark.
- ▶ Combining IS with the policy rule gives the **AD curve**:  $Y - \bar{Y} = Z - a(\pi - \pi^*)$ .
- ▶ Changes in  $\pi$  generate **movements along** AD; changes in  $G$ ,  $T_0$ ,  $v$ , or  $\hat{p}$  **shift** AD.
- ▶ When standard transmission weakens, **unconventional policy** works through the spread  $\hat{p}_t$  and longer-term yields.

## Preview: Lecture 5

Next time you add the supply side by studying the **Phillips curve** and the **AS curve** in Chapter 18.

- ▶ The AD curve tells you how inflation affects **demand** through monetary policy.
- ▶ The Phillips curve tells you how output and unemployment affect **inflation**.
- ▶ Together they form the AS-AD model used throughout the next block of the course.

# Appendix

## Appendix: Exact Derivation of the IS Benchmark

Start from the linear goods-market model

$$Y = C + I + G, \quad C = C_0 + cY^d, \quad Y^d = Y - T, \quad T = T_0 + tY, \quad I = I_0 - br.$$

Insert  $T$  into disposable income:

$$Y^d = Y - (T_0 + tY) = -T_0 + (1 - t)Y.$$

Then consumption becomes

$$C = C_0 + c[-T_0 + (1 - t)Y] = C_0 - cT_0 + c(1 - t)Y.$$

Insert  $C$  and  $I$  into equilibrium output:

$$Y = [C_0 - cT_0 + c(1 - t)Y] + [I_0 - br] + G.$$

Collect the  $Y$  terms:

$$[1 - c(1 - t)]Y = C_0 - cT_0 + I_0 - br + G.$$

## Appendix: Exact Derivation of the IS Benchmark, continued

Divide by  $1 - c(1 - t)$ :

$$Y = \frac{1}{1 - c(1 - t)} (C_0 - cT_0 + I_0 - br + G).$$

Define

$$\alpha_1 \equiv \frac{1}{1 - c(1 - t)}, \quad \alpha_2 \equiv \frac{b}{1 - c(1 - t)}, \quad \alpha_3 \equiv \frac{c}{1 - c(1 - t)},$$
$$V \equiv \frac{C_0 + I_0}{1 - c(1 - t)}.$$

Then

$$Y = \alpha_1 G - \alpha_2 r - \alpha_3 T_0 + V.$$

Subtract the structural counterpart

$$\bar{Y} = \alpha_1 \bar{G} - \alpha_2 \bar{r} - \alpha_3 \bar{T}_0 + \bar{V}$$

to obtain

$$Y - \bar{Y} = \alpha_1 (G - \bar{G}) - \alpha_2 (r - \bar{r}) - \alpha_3 (T_0 - \bar{T}_0) + v, \quad v \equiv V - \bar{V}$$





## Appendix: Exact and Approximate Real Interest Rates

The exact ex ante real rate satisfies

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e} \iff r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e} - 1.$$

With  $i_t = i_t^p + \hat{\rho}_t$  and the Taylor rule

$$i_t^p = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + \beta(Y_t - \bar{Y}),$$

you have the exact mapping

$$r_t = \frac{1 + \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + \beta(Y_t - \bar{Y}) + \hat{\rho}_t}{1 + \pi_{t+1}^e} - 1.$$

If inflation and nominal interest rates are moderate, this becomes the familiar shortcut

$$r_t \approx \bar{r} + h(\pi_t - \pi^*) + \beta(Y_t - \bar{Y}) + \hat{\rho}_t$$

used in the chapter's AD derivation.

## Appendix: Finishing the AD Derivation

Start from

$$\begin{aligned}Y - \bar{Y} &= \alpha_1(G - \bar{G}) - \alpha_2(r - \bar{r}) - \alpha_3(T_0 - \bar{T}_0) + v, \\r - \bar{r} &= h(\pi - \pi^*) + \beta(Y - \bar{Y}) + \hat{\rho}.\end{aligned}$$

Insert the real-rate rule into IS:

$$\begin{aligned}Y - \bar{Y} &= \alpha_1(G - \bar{G}) - \alpha_2[h(\pi - \pi^*) + \beta(Y - \bar{Y}) + \hat{\rho}] - \alpha_3(T_0 - \bar{T}_0) + v \\&= \alpha_1(G - \bar{G}) - \alpha_2h(\pi - \pi^*) - \alpha_2\beta(Y - \bar{Y}) - \alpha_2\hat{\rho} - \alpha_3(T_0 - \bar{T}_0) + v.\end{aligned}$$

Move the endogenous output-gap term to the left:

$$(1 + \alpha_2\beta)(Y - \bar{Y}) = \alpha_1(G - \bar{G}) - \alpha_3(T_0 - \bar{T}_0) + v - \alpha_2h(\pi - \pi^*) - \alpha_2\hat{\rho}.$$

## Appendix: Finishing the AD Derivation, continued

Now divide by  $(1 + \alpha_2\beta)$ :

$$Y - \bar{Y} = \frac{\alpha_1(G - \bar{G}) - \alpha_3(T_0 - \bar{T}_0) + v - \alpha_2\hat{p}}{1 + \alpha_2\beta} - \frac{\alpha_2 h}{1 + \alpha_2\beta}(\pi - \pi^*).$$

Define

$$Z \equiv \frac{\alpha_1(G - \bar{G}) - \alpha_3(T_0 - \bar{T}_0) + v - \alpha_2\hat{p}}{1 + \alpha_2\beta}, \quad a \equiv \frac{\alpha_2 h}{1 + \alpha_2\beta}.$$

Then

$$Y - \bar{Y} = Z - a(\pi - \pi^*)$$

which is the AD curve used in the rest of the course.



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